Development Proposal Medical Device Interface Communication and Integration

Submitted by: eSpark Consulting Group
Role: Medical Device Interface Communication and Integration Specialist
Hourly Rate: $25/hour
Availability: 20–30 hours per week
Date: 14/5/2025

Overview

I am submitting this proposal in response to BMR TechWorks' call for a Medical Device Interface Communication and Integration Specialist. With experience in healthcare interoperability and a strong technical background in device protocols and middleware integration, I am well-prepared to contribute to your mission of advancing real-time data exchange and clinical workflow efficiency.

Scope of Services:

Based on the provided job overview and the M2 Communication Protocol document, I propose the following scope of work:

1. Device Interface Development

- Implement TCP/IP-based communication following the M2 protocol over ports 1401 and 1402.
- Establish bi-directional command handling for monitor data, logs, status, settings A/B/C, and patient information.
- Utilize the defined packet structure: Header → Data → Checksum → Delimiter.

2. Protocol Parsing & Handling

- Parse and respond to commands: ACK, Status, Monitor Data, Log, and Settings Write commands.
- Implement AES encryption and MD5 integrity verification for secure data parsing.
- Integrate retry and error handling logic based on end codes and timeouts.

3. Data Integrity & Queue Management

- Build a concurrent session management system to handle multiple device streams.
- Ensure robust communication handling, including retransmissions and fallback mechanisms.

4. Real-Time Monitoring Dashboard

- Develop an interface for real-time monitoring of:
    - Active device sessions
    - Monitor data streams
    - Log messages with timestamps and device identifiers
    - Session and encryption status

5. Documentation & Handoff

- Provide complete integration documentation including:
    - Protocol mappings
    - Workflow diagrams
    - Setup and deployment instructions
- Deliver clean, reusable, and tested integration code.

Technical Stack & Tools

- **Programming**: C#, Background services, .NET 6/7
- **Protocols**: HL7, DICOM, TCP/IP, BLE, Serial, Custom Protocols (per M2 spec)
- **Tools**: Visual Studio, Wireshark, Postman, Git, Docker (if needed)
- **Database**: SQL Server, PostgreSQL (for logs/patient/session tracking)
- **Security**: AES encryption, MD5 integrity check per M2 guidelines

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Deliverables Per Device Integration:

Phase	Description	Output
Initial Device Review	Understand protocol and verify communication	Successful handshake and protocol test
Interface Implementation	Build command handling, status responses, encryption logic	Integration module
Queue System Setup	Session handling, timeout, and retry logic	Concurrent connection manager
Monitoring Dashboard	UI/dashboard to view data and logs	Web panel or middleware component
Documentation & Delivery	Final delivery of working code with notes	README, tech docs, source code

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Pricing Model:

- **Rate**: $25/hour
- **Weekly Cap**: 20–30 hours

Conclusion:

I am confident that my experience in medical device communication, protocol parsing, and secure data integration aligns closely with BMR TechWorks' project requirements. By implementing a reliable, encrypted interface using the M2 protocol, along with real-time monitoring and robust session management, I aim to ensure seamless, secure, and efficient data exchange across devices. With a strong technical foundation and commitment to clean, well-documented code, I am prepared to contribute meaningfully to your team and support the timely and successful delivery of this integration project.